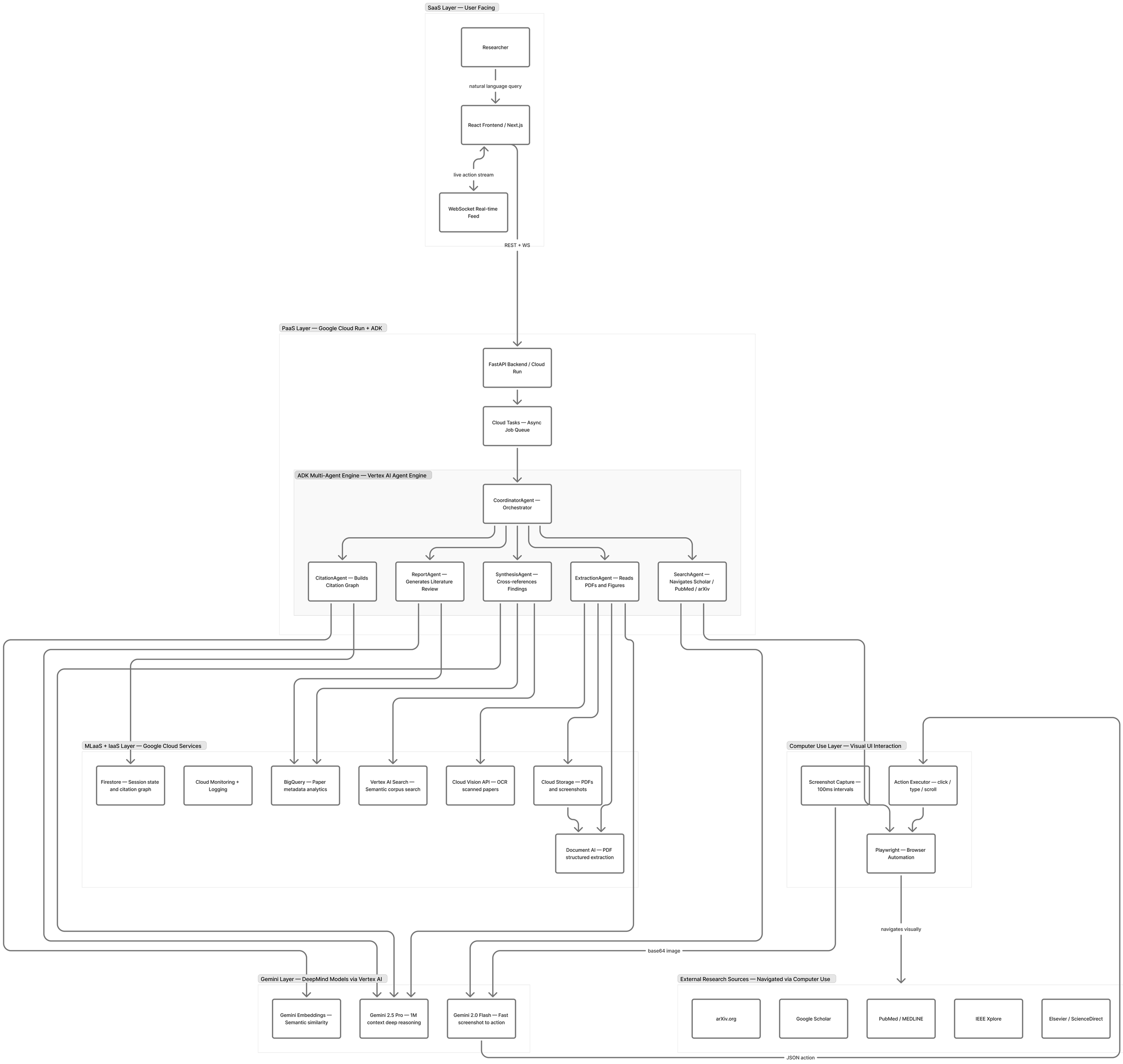
SaaS Layer — User Facing
Researcher
natural language query
React Frontend / Next.js
live action stream
WebSocket Real-time Feed
REST + WS
PaaS Layer — Google Cloud Run + ADK
FastAPI Backend / Cloud Run
Cloud Tasks — Async Job Queue
ADK Multi-Agent Engine — Vertex AI Agent Engine
CoordinatorAgent — Orchestrator
CitationAgent — Builds Citation Graph
ReportAgent — Generates Literature Review
SynthesisAgent — Cross-references Findings
ExtractionAgent — Reads PDFs and Figures
SearchAgent — Navigates Scholar / PubMed / arXiv
MLaaS + IaaS Layer — Google Cloud Services
Firestore — Session state and citation graph
Cloud Monitoring + Logging
BigQuery — Paper metadata analytics
Vertex AI Search — Semantic corpus search
Cloud Vision API — OCR scanned papers
Cloud Storage — PDFs and screenshots
Document AI — PDF structured extraction
Computer Use Layer — Visual UI Interaction
Screenshot Capture — 100ms intervals
Action Executor — click / type / scroll
Playwright — Browser Automation
navigates visually
base64 image
Gemini Layer — DeepMind Models via Vertex AI
Gemini Embeddings — Semantic similarity
Gemini 2.5 Pro — 1M context deep reasoning
Gemini 2.0 Flash — Fast screenshot to action
External Research Sources — Navigated via Computer Use
arXiv.org
Google Scholar
PubMed / MEDLINE
IEEE Xplore
Elsevier / ScienceDirect
JSON action